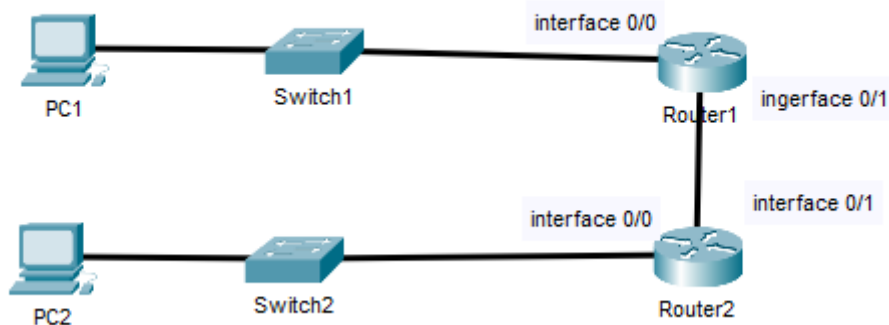


Lab 4: Building a Switch and Router Network on both IPv4 and IPv6

Topology



Step 0: useful commands

Router#show ip int brief	show a list of the routers network interfaces. You can find the names and status of all interfaces here
Router#write erase	delete the startup config
Router#reload	reboot the router
Router(config)#interface <NameOfInterface>	navigate to the configuration mode of an interface
Router(config-if)#ip address <IPAddress> <Subnetmask>	Set the IP address and subnet mask of a router's interface
Router(config-if)#no shutdown	Enable the interface
Router(config)#ip route <destination network address> <destination network subnet mask> <ip address of next router>	Configure a static route
Router#show ip route	Show router's routing table
Router#show interface Gigabitethernet 0/0	Show information of a specific router interface
Router(config)#ipv6 unicast-routing	Enable SLAAC on the router

FOR THIS LAB, YOU CAN WORK ALONE OR IN A GROUP OF 2.

Step 1: Subnet the network

To randomize the IP addresses that each group uses, you will use the last 2 digits of your student number in the IP network address that you can use for your network (if you work in a group, choose one student number). The network that you can use to subnet the network is 10.10.(student card digits).0/22.

e.g. for the student number r215468, the network address becomes 10.10.68.0/22

We will use: 10.10.9.0/22

Use this address to subnet the network given above and assign IP addresses to each device in the network. Make sure to maximize the number of IP addresses on each network.

Let the table be checked before going to the next step

Assigned IP addresses:

Device	Interface	IP Address	Subnet Mask	Default Gateway (if applicable)
Router1	0/0			
	0/1			
Router2	0/0			
	0/1			
PC1				
PC2				

Step 2: Prepare your network in packet tracer

- Before building a network on physical devices, it is always interesting to simulate your network first. Building a virtual network is quicker, you can find problems easier (in simulation mode) and this way you can prepare all the commands that you need in the physical network that you know will work.
- Build the network as shown in the

IPv4

- Assign IP addresses to the PC's, switches and routers.
- Before testing the network connectivity, think about how PC's and routers work and predict which devices can reach each other with the current configuration.
- Test network connectivity between all the devices and see if your prediction was correct. You can also use the simulation mode to see what is happening.
- PC1 cannot reach PC2, PC2 cannot and PC1. The reason is that the subnet of PC2 is not yet known to Router1 and the subnet of PC1 is not yet known to Router2. Have a look at the routing table of both routers to confirm.
- Configure a static route on Router1 to the subnet of PC2, and a static route on Router2 to the subnet of PC1. Confirm the configuration in the router's routing table.
- Test network connectivity between PC1 and PC2.

IPv6

- Only PC1 and Router1 have to be configured with IPv6. In IPv6, you don't need to configure static IP addresses to the PC's. SLAAC (Stateless Address Autoconfiguration) enables PC's connected to an IPv6 router to automatically configure an IP address in the correct network.
- Configure the IPv6 address 2001:A:CAFE:1::1/64 to Router1 and enable IPv6 unicast routing.
- PC1 should automatically get an IPv6 address
- Test network connectivity between PC1 and Router1 over IPv6

Step 3: Building the physical network

- Cable the physical network as shown in the topology
- During this lab, you will only have to configure the router. However, you should reset your switches and routers first.

IPv4

- Configure the network for IPv4
 - o Assign names to the routers
 - o Assign IP addresses to all the devices
 - o Configure static routes on Router1 and Router2
- Test network connectivity between all the devices

IPv6

- Configure PC1 and Router1 for IPv6
 - o Static IP address on Router1
 - o Enable IPv6 unicast routing on Router1
 - o PC1 should receive an IPv6 address automatically (SLAAC)
- Test network connectivity between PC1 and Router1

Step 4: Monitor the network for IPv4 traffic

- Start monitoring with Wireshark **on both computers** and ping over IPv4 from PC1 to PC2.
 - o The Ping requests travel from PC1 to PC2. The same packet can be seen in Wireshark on both computers. On PC1, the ping request will leave the interface while the same packet will arrive in PC2. Fill in the table to see what changed in the packet

	Source MAC	Destination MAC	Source IP	Destination IP
Ping request (traveling from PC1 to PC2) as seen in Wireshark on PC1				
Ping request (traveling from PC1 to PC2) as seen in Wireshark on PC2				

- Find out to which devices the mac addresses belong and add the name of the device to the table above.
- Why are the mac addresses not the same? What happened with these mac addresses at which point in the network?
- Why are the ip addresses the same?

Demo

After you did the lab, you should be able to demonstrate the following. Think about how you can demonstrate these achievements.

- Router name
- Network connectivity between PC1 and Router1 over IPv4
- IPv6 SLAAC address on PC1
- Network connectivity between PC1 and Router1 over IPv6
- Network connectivity between PC0 and PC1 over IPv4
- Explain the routers routing table
- Source and Destination MAC and IP address in Wireshark capture on PC0 and PC1